

Outline & General Information

MODELING OF MULTI-PHYSICS PHENOMENA

BIONICS ENGINEERING - A.A. 2025-26

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General informations

- Teachers
 - Prof. Antonio De Simone - Scuola Superiore Sant'Anna Pisa
 - Prof. Alessandro Tognetti – Università di Pisa
 - Coteacher
- Main aim: learn how to model and solve bionics-relevant physical and multi-physical problems
 - Analytical and numerical solutions (finite elements, finite differences)
- Exam: oral exam, theoretical and “practical” questions
- Reference Book: “Modelling Organs, Tissues, Cells and Devices Using MATLAB and COMSOL Multiphysics”
Dokos

Main Topics

- Part 1 (prof De Simone)
 - Elements of fluid dynamics
 - Laminar flows (Navier-Stokes equations)
 - Creeping flows (Stokes equations)
 - Elements of 3D elasticity
 - Linear elasticity
 - Topics in nonlinear elasticity
- Part 2 (Prof Tognetti)
 - Modeling bioelectricity, neural models and BIO-Tissues stimulation
 - Modeling bioinspired sensors
 - Numerical solution methods (FE, FD)
 - Optional: project work
 - Implement solutions from literature examples